

Melroy Dcosta

M.Sc. Data Science | Data Analytics | Business Intelligence | Applied Machine Learning

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Marburg, Germany

Profile

M.Sc. Data Science student at Philipps-Universität Marburg with hands-on experience transforming complex datasets into actionable insights using Python, SQL, Excel and data visualization. Experienced in data cleaning, exploratory analysis, KPI reporting, dashboard development, predictive modeling and analytical workflow automation. Built end-to-end analytics projects involving **541,000+ retail transactions** and large-scale energy datasets, combining strong technical analysis with business-oriented problem solving.

Technical Skills

Data Analytics & BI: EDA, Statistical Analysis, Data Cleaning, Data Transformation, KPI Analysis, Dashboard Development, Data Visualization, Power BI, Microsoft Excel, Streamlit, Plotly

Programming & Data: Python, SQL, R, Pandas, NumPy, SQLite, Data Pipelines, Data Validation, Data Profiling

Machine Learning: Scikit-learn, XGBoost, PyTorch, TensorFlow, Predictive Modeling, Feature Engineering, Hyperparameter Optimization, Model Evaluation, SHAP

AI & Engineering: NLP, RAG, LangChain, LangGraph, REST APIs, Flask, Docker, Git/GitHub, Linux, Jupyter, Workflow Automation

Experience

Data Analyst Intern

Apr. 2024 – Aug. 2024

Indian Institute of Technology (IIT) Roorkee

India

- Analyzed, cleaned and structured datasets using Python and SQL; performed exploratory analysis, data profiling and validation to transform raw information into reliable analytical outputs.
- Developed visualizations and analytical reports to identify trends, KPIs and actionable insights, while documenting methods and findings for reproducible data-driven decision making.

AI / Machine Learning Intern

Jun. 2023 – Dec. 2023

Zephyr Technologies

Mangalore, India

- Processed and modeled **50,000+ records**; applied feature engineering and hyperparameter optimization, improving predictive model accuracy by approximately **20%**.
- Developed Flask-based inference APIs and monitoring/retraining workflows, reducing prediction error by approximately **15%** and supporting integration of ML models into application workflows.

Education

Philipps-Universität Marburg

Oct. 2024 – Present

M.Sc. Data Science

Marburg, Germany

St Joseph Engineering College

Jul. 2020 – May 2024

B.E. Artificial Intelligence & Machine Learning

India

Projects

E-Commerce Analytics Dashboard

Python, SQL, Streamlit, Plotly

- Built an end-to-end analytics solution for **541,000+ retail transactions**, combining ETL, SQL data marts, data-quality validation, RFM segmentation and interactive dashboards for revenue, orders, customers, products and returns.

Smart Energy Analytics

Python, Data Analytics, Data Engineering

- Built a scalable energy-data analytics workflow to process the complete **112-block dataset**, applying memory-efficient data processing and storage techniques to prepare large-scale energy data for analysis, trend exploration and analytical reporting.

Restaurant Success Prediction

Python, XGBoost, SHAP

- Developed predictive models achieving approximately **85% accuracy**; combined EDA, feature engineering and explainability techniques to identify important business and performance drivers.

RAG-Based PDF Knowledge Assistant

Python, LangChain, LangGraph

- Developed a document analytics and question-answering system using PDF processing, embeddings, vector retrieval and LLM-based generation with FAISS/Chroma and LangChain/LangGraph workflows.

Deep Learning Image Steganography

Python, PyTorch, Computer Vision

- Built a deep-learning image steganography workflow for encoding and recovering hidden visual information and evaluated reconstruction quality and robustness across image-processing experiments.

Certifications & Additional Information

Certifications: IBM Data Science Professional Certificate; Introduction to Supply Chain Finance & Blockchain Technology

Languages: English – Fluent | German – A2, actively improving

Data Engineering & Analytics: Proven ability to build Python-based analytical and AI workflows, process large datasets and translate complex technical data into clear, actionable insights.